

Spec Sheet Verification v0

Final engineering report covering the verification pipeline, experiments, winning architecture, measured results, and remaining manual-review boundary.

77%

FINAL VERIFIED

23%

HUMAN REVIEW

0

FINAL FAILED

110

TESTS PASSING

Final validated aggregate: 77 verified / 23 human review / 0 failed across the 100-PDF benchmark after provider-failure recovery. The clean end-to-end run exposed four transient Gemini failures; retry/rescue handling was then added and the four cases were recovered or safely downgraded to human review.

From raw PDFs

Clean pipeline cost

\$0.2721 total for the 100-PDF full run before the two-document live recovery. The final live recovery added **\$0.00146**.

Conservative by design

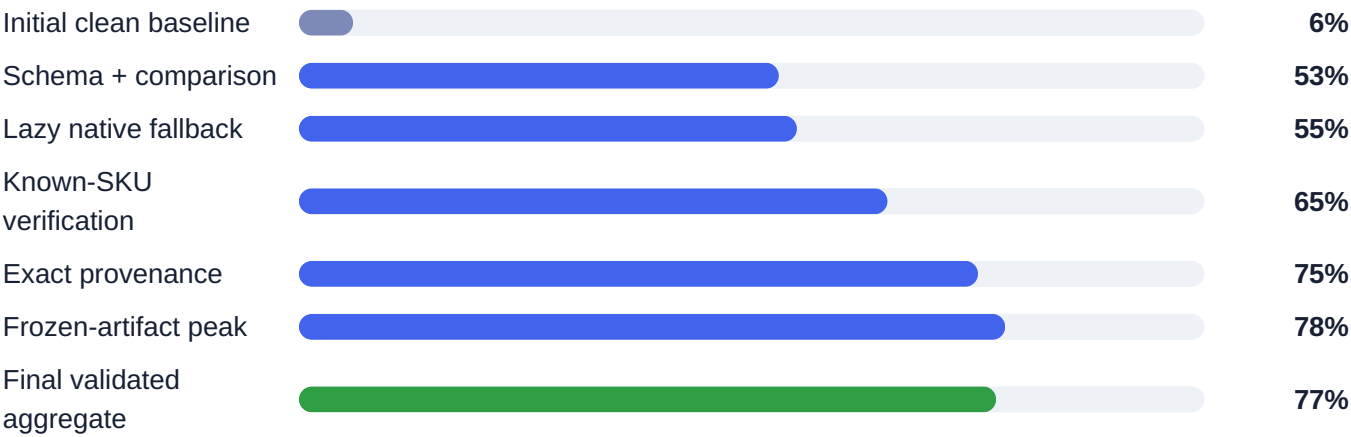
What “verified” means

Automatic approval only when independent content evidence, known-SKU evidence, or exact source provenance is strong enough. Otherwise the document remains human review.

Prepared for Reuben
BuilderLab - 18 September 2026

What changed the result

The largest gains did not come from adding a stronger OCR engine. They came from correcting the verification problem itself: use the PDF content when it is sufficient, use the known upstream SKU when identity is already known, and use exact manufacturer provenance when a shared or portfolio PDF cannot prove a unique SKU internally.



Worked

Semantic verification

Part-number normalization and less brittle description comparison removed large numbers of false mismatches.

Worked

Known SKU as a claim

`index.csv` already maps every benchmark PDF to an upstream SKU, so the pipeline verifies that claim instead of always rediscovering identity.

Worked

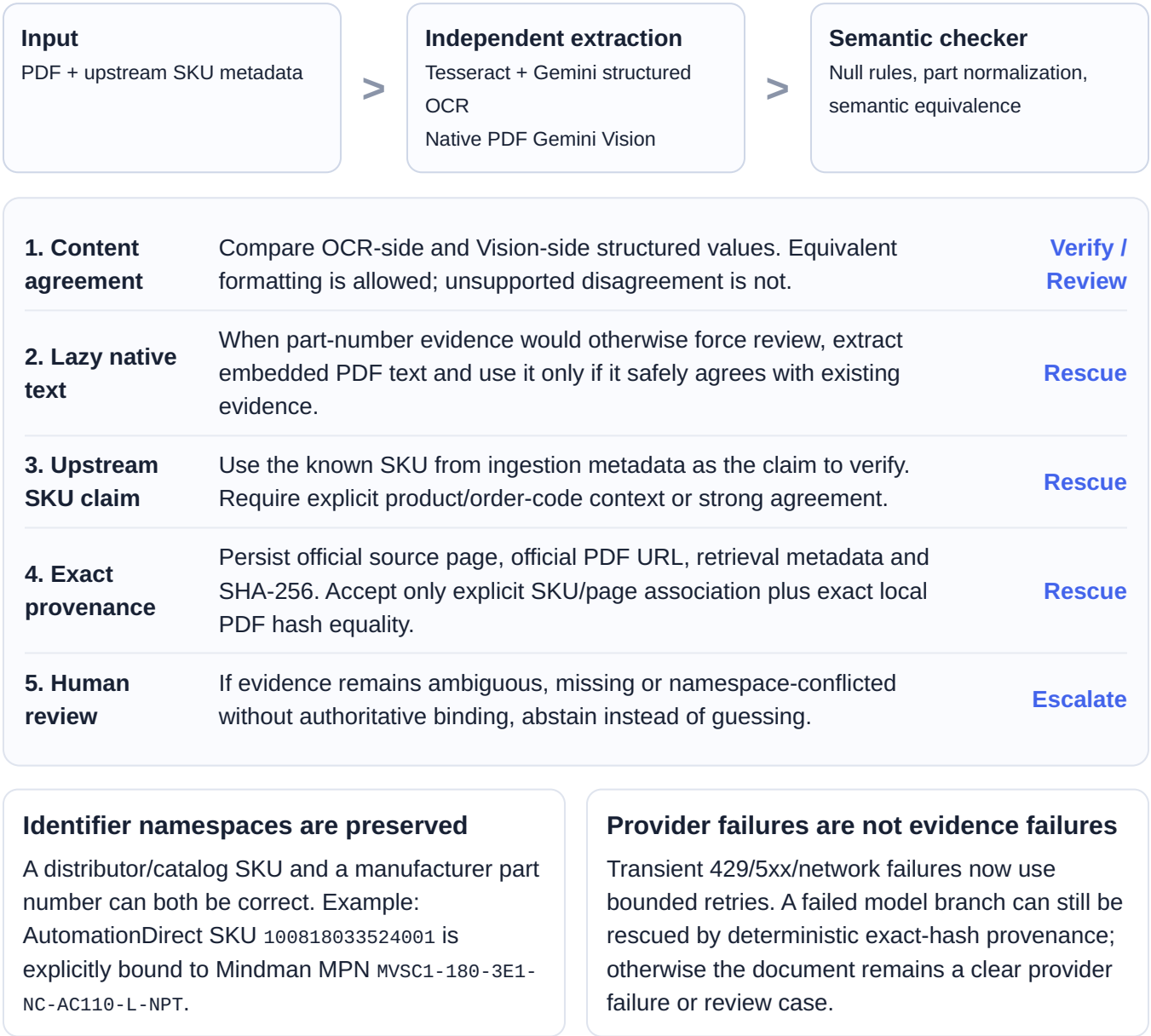
Exact provenance

Official product/revision page -> official PDF -> exact local SHA-256 match safely rescued shared/family datasheets.

Core lesson: the difficult documents were often not unreadable. They were documents where a unique orderable SKU was absent, ambiguous, or represented in a different identifier namespace. Verification needed more than OCR.

Multi-signal verification, with conservative abstention

The final v0 keeps independent extraction paths, deterministic evidence, and source provenance separate. No path is allowed to invent or “correct” a value.



What worked, and what we rejected

Approach	Outcome	Decision
Schema clarification + semantic comparison	Raised the benchmark from 6% to 53% by separating true mismatch from punctuation, units and compatible description detail.	Keep
Lazy native PDF text fallback	Recovered difficult part-number cases without paying the native-text cost on every document.	Keep
Upstream SKU claim verification	Changed the problem from “discover the SKU” to “prove the known SKU”. Reached 65%.	Keep
Exact external provenance	Manufacturer/product page -> linked PDF -> exact SHA-256 safely handled shared/family datasheets and namespace bindings.	Keep
Provider retry + branch tolerance	Eliminated transient 503s as terminal document failures and allowed deterministic provenance rescue.	Keep
Anchor-bounded OCR / structured anchor context	Low recovery and regressions; geometry was not stable enough across layouts.	Reject
PaddleOCR alternative engine	CPU-heavy and no useful completed benchmark gain.	Reject
Docling layout/table parsing	45/45 parsed but candidate recall was worse than the existing evidence-first method; no missing cases recovered.	Reject
Evidence-first candidate ranking	Safe but low recall; the likely-correct candidate was absent from most difficult cases under the evaluation proxy.	Not integrated
KEB provenance recovery	Family and size evidence existed, but no explicit full-SKU -> exact-document chain. 0 safe rescues.	Stop

Deliberate constraint: no filename-only verification, no fuzzy PDF matching, no family-page assumption by itself, and no automatic promotion when the evidence is merely plausible.

Measured v0 performance

Clean end-to-end run

75 verified / 21 review / 4 failed

The four failures were transient Gemini provider errors, not verification-logic failures. The clean run cost **\$0.2720724** across 100 PDFs.

Average 27.90 s/document; median 8.53 s/document.

After reliability recovery

77 verified / 23 review / 0 failed

Two failures were deterministically recovered through accepted exact-hash provenance. The two remaining provider-failed PDFs were rerun live and correctly became human review.

Live two-document recovery cost: \$0.0014624.

Metric	Final / measured value
Final automatic verification rate	77%
Final human-review rate	23%
Final failed rate after recovery	0%
Full clean-run API cost	\$0.2721 / 100 PDFs
Approx. API cost per PDF in clean run	\$0.00272
Tests	110 passing
Resumability	Passed
Network during persisted provenance evaluation	0 requests

Why 78% became 77% in the final aggregate

The 78/22/0 result was measured against frozen extraction artifacts. A fresh end-to-end run exposed normal model stochasticity: 04A-B01 moved from verified to human review. We therefore use **77/23/0** as the more honest final v0 result.

Reliability condition achieved: no document remains failed because of provider availability. Uncertain evidence is routed to human review, while deterministic provenance remains available as an independent rescue path.

What remains manual, and why v0 is done

The remaining 23% is dominated by documents where the exact orderable SKU is absent from the PDF and no production-safe external mapping exists. These are not parser crashes. They are evidence-boundary cases.

Main unresolved groups

- **KEB portfolio documents:** shared portfolio PDF, family/size evidence only, no explicit full-SKU chain.
- **BRECOflex timing belts:** series/type information exists, but complete order-code mapping is not proven.
- **dPOFLEX family sheets:** shared family document with multiple variants and no exact SKU binding.
- **Other ambiguous cases:** exact SKU not present or not authoritative enough to promote automatically.

Correct v0 behavior

- Auto-verify when evidence is strong and auditable.
- Preserve separate identifier namespaces.
- Use persisted source provenance offline.
- Retry transient provider failures.
- Abstain to human review when proof is insufficient.

v0 release evidence to freeze

Clean full-run report

`.runs/evaluation-100-spec-sheet-clean-final-20260918-v2/report.json`

Clean full-run manifest

`.runs/evaluation-100-spec-sheet-clean-final-20260918-v2/manifest.json`

Live provider-recovery report

`.runs/provider-failure-live-20260918/report.json`

Recommendation: ship v0 at 77% automatic verification with human review as the explicit safety boundary. Further automation should be source-specific provenance work, not more generic OCR experimentation, and only if product requirements justify the extra effort.

Suggested release label: v0

Suggested release commit: `release: freeze spec sheet verification v0`